Scenario S2-O1-I1 — Our Level 1 (Bootstrapped) / Investor Level 1 (Bootstrapped)

Scenario Summary

Maximum-constraint path. The founders self-fund at bootstrapped intensity (~40% of ideal spend) through Pre-Pilot and Pilot 1, then a bootstrapped-level investor arrives during the Pilot 1 to Pilot 2 transition (approximately M7-M9). Both parties operate at bare-minimum budgets throughout, creating the longest timeline and most sequenced execution of any S2 combination.

- **Economic cost to Production:** ~$385,400 / INR 31,988,200
- **Target timeline to Production:** 30 months
- **Founder/company capital before investor:** ~$115,700 / INR 9,603,100
- **Investor capital modeled:** ~$133,000 / INR 11,039,000
- **Equity structure:** 60 / 20 / 20 (founder / founding team / investor pool)
- **Main risk to watch:** extended runway creates fatigue risk; bootstrapped investor capital barely covers Pilot 2 + Production staffing ramp

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Timeline

Phase	Timing	Duration	Minor Checkpoints	Gate to Advance
Pre-Pilot	M0-M6	6 months	Architecture freeze (M2), APK backlog lock (M3), doctor training draft (M4), admin callback SOP (M5), internal rehearsal cycles (M5-M6).	Core web flows stable at reduced scope, internal rehearsals pass, doctor and admin runbooks versioned.
Pilot 1	M6-M9	3 months	Warm-network onboarding, first paid consults, callback calibration, doctor playbook iteration, investor pitch deck finalized.	Ethiopia pilot runs live for 2-3 months with measured throughput; investor term sheet signed.
Pilot 2	M9-M18	9 months	Kenya prep (M12), partner reporting (M14), roster planning, partial automation, staffing calibration, refresher training.	5k-10k user-capable operations with stronger APK, cleaner reporting, staffed callback coverage.
Production Readiness	M18-M30	12 months	Compliance closeout (M22), release freeze (M25), launch playbooks (M27), certification refreshers, production support schedule.	Current prototype scope is production-safe across web and patient mobile with documented fallback operations.

Feature and Output Checklist

Phase	Output Checklist
Pre-Pilot	Web core hardened at reduced pace; APK build underway (sequenced, not parallel); doctor/admin training materials drafted; support desk scripts and callback trees defined. Slower iteration due to constrained hours.
Pilot 1	Patient app covers core journeys (registration, dashboard, basic consult); provider web is stable for GP flow; doctor roster, admin callbacks, and manual fulfilment visible in-system. Feature set narrower than L2/L3 — specialist and diagnostics flows deferred.
Pilot 2	Core patient features complete; partner dashboards functional; staffing ladder active at minimum levels; reporting and payment loops operational but manual-heavy. Kenya prep begins late in phase.
Production Readiness	Current prototype scope is production-grade across web and patient mobile. Trained ops staff with manual fallback available. Compliance documentation complete. Extended timeline allows thorough but slow hardening.

Services and SLAs

Phase	Uptime Target	Response Commitment	Operating Model
Pre-Pilot	95.0%	Next business day	3 core team at reduced utilization + 7 part-time developers at ~40% capacity + 2 advisors; no employed clinical coverage, only training and rehearsals.
Pilot 1	96.5%	Same-day for critical issues	Minimum employed coverage: 2 doctors, 2 admin ops, 2 technical support. Friendly clinics supplement.
Pilot 2	97.5%	4-hour P1 response	Minimum employed coverage: 4 doctors, 4 admin ops, 2 technical support. Callback and partner reporting loops active.
Production	98.5%	2-hour P1 response	Minimum employed coverage: 6 doctors, 6 admin ops, 3 technical support for near-24/7 coverage.

Cost Breakdown

Summary

Category	Amount
Capital expenditure through Production	$172,300 / INR 14,300,900
Operating expenditure before Production launch	$213,100 / INR 17,687,300
Total economic spend to Production	**$385,400 / INR 31,988,200**

Workstream Detail

Workstream	Economic Cost (USD)	Economic Cost (INR)	Notes

Product and founder office	$95,040 /	INR 7,888,320	3 core team at $20/hr x ~88 hrs/mo x 40% utilization-adjusted = ~$3,520/mo for 27 productive months (phased). Remaining months at higher utilization post-investor.
Web platform hardening	$14,900	INR 1,236,700	745 hours at $20/hr senior rate. Stretched over longer timeline due to constrained hours.
Patient APK build	$7,200	INR 597,600	Fixed Android quote: 480 hours at $15/hr. Sequenced rather than parallel with web work.
QA, DevOps, security	$8,800	INR 730,400	~440 hours at $20/hr. Reduced scope; minimum viable hardening.
Admin and care operations	$119,710	INR 9,935,930	Employed doctors, admin callback staff, and technical support ladder across 24 months of live operations. See ops staffing breakdown below.
Partner onboarding and provider success	$8,250	INR 684,750	Doctor/admin training protocols, SOP writing, partner reporting. Lean execution.
Legal, accounting, compliance	$4,800	INR 398,400	Friendly-network pricing at economic value.
Infrastructure, phones, and tools	$28,700	INR 2,382,100	Hosting, devices, telecom/call costs, operational tooling across 30 months. Starts minimal, scales with phases.
Marketing, travel, rehearsals	$8,000	INR 664,000	Bare minimum. Word-of-mouth and warm-network driven.
Contingency and buffer	$90,000	INR 7,470,000	~23% buffer on total. Essential given bootstrapped constraints and long timeline.

Operational Staffing Cost by Phase

Phase	Duration	Monthly Ops Cost (USD)	Monthly Ops Cost (INR)	Phase Total (USD)
Pre-Pilot (M0-M6)	6 months	$0	INR 0	$0
Pilot 1 (M6-M9)	3 months	$2,482	INR 206,006	$7,446
Pilot 2 (M9-M18)	9 months	$4,361	INR 361,963	$39,249
Production Readiness (M18-M30)	12 months	$6,542	INR 542,986	$78,504
Total operational staffing				**$125,199**

Infrastructure Cost by Phase

Phase	Duration	Monthly Infra (USD)	Monthly Infra (INR)	Phase Total (USD)
Pre-Pilot	6 months	$80	INR 6,640	$480
Pilot 1	3 months	$300	INR 24,900	$900
Pilot 2	9 months	$550	INR 45,650	$4,950
Production Readiness	12 months	$900	INR 74,700	$10,800
One-time equipment (phones, laptops)	—	—	—	$1,200
Total infrastructure				**$18,330**

Effort and Team

Function	Staffing Pattern	Notes
Core team	3 part-time members @ ~4 hrs/day each, ~40% effective utilization pre-investor	Single owner/founder + 2 founding members. Heavy sequencing of tasks.
Technical delivery (pre-investor)	7 part-time developers, modeled at ~200 productive hrs/mo	~40% of ideal 480 hrs/mo. High sequencing, minimal parallelism.
Technical delivery (post-investor)	7 part-time developers, modeled at ~240 productive hrs/mo	Investor capital enables slightly higher utilization (~50% of max). Still constrained.
Advisory board	2 fractional advisors @ ~10-15 hrs/mo each	Finance, compliance, strategic review.
Employed care coverage	Pilot 1: 2 doctors / Pilot 2: 4 / Production: 6	Partner clinics still critical at bootstrapped levels.
Employed admin ops	Pilot 1: 2 admins / Pilot 2: 4 / Production: 6	Callbacks, exception handling, manual fulfilment.
Technical support	Pilot 1: 2 / Pilot 2: 2 / Production: 3	L1/L2 issue triage and live-ops monitoring.

Monthly Team Economic Cost

Period	Dev Team (USD)	Dev Team (INR)	Notes
Pre-investor (M0-M8)	~$5,930	~INR 492,190	~40% of full $14,830/mo rate card. Bare minimum.
Post-investor (M9-M30)	~$7,400	~INR 614,200	~50% of full rate card. Investor enables modest increase.

Revenue and Pricing

Revenue Source	Pilot 1	Pilot 2	Production
GP consults	Subsidised, Ethiopia-first, ~$2-3 per paid interaction	Full Ethiopia pricing + Kenya tests	Core revenue line
Specialist consults	Minimal	Grows slowly after referral flow stabilizes	Lower-volume margin contributor
Pharmacy commission	Small, admin-assisted	Repeatable, manual-heavy	Scales with partner conversion
Diagnostics commission	Small, admin-assisted	Improves with order flow consistency	Retention lever
Travel/care coordination	Rare, high-touch	Low-volume, higher-value	Strategic differentiator

Revenue Line	Ethiopia Benchmark	Kenya Benchmark	Notes
GP consult	$3.5 / ETB 470	$5 / KES 650	Starts subsidised in Pilot 1; reaches benchmark by late Production.
Specialist consult	$9 / ETB 1,250	$12 / KES 1,560	Pilot 2 onward.
Pharmacy take-rate	10% of order value	10% of order value	Pilot 1 onward with manual fulfilment.
Diagnostics take-rate	12% of order value	12% of order value	Pilot 1 onward with admin/lab coordination.
Travel/care coordination	$10-15	$10-15	Small volume, high-touch.
Subscription	Deferred	Deferred	Not modeled before meaningful scale.

CAC and Key Business Metrics

Metric	Pilot 1	Pilot 2	Production
CAC	$10	$16	$22
Gross margin	35%	45%	55%
Monthly ARPU	$1.5	$2.5	$4
Monthly churn	15%	12%	8%
LTV/CAC target	1.2x	2.0x	2.8x

Cash Flow and Runway

Phase Summary

Checkpoint	Ending Cash (USD)	Ending Cash (INR)	Approx. Runway Remaining	Trigger Threshold
Pre-Pilot (M6)	$11,400	INR 946,200	1.2 months	Trigger contingency if cash drops below 2 months of current burn.
Pilot 1 (M9)	$8,200	INR 680,600	0.7 months	Critical: investor close must happen by M9.
Pilot 2 (M18)	$62,800	INR 5,212,400	4.5 months	Trigger contingency if cash drops below 3 months of burn.
Production Readiness (M30)	$71,500	INR 5,934,500	4.2 months	Trigger contingency if cash drops below 3 months of burn.

Monthly Cash Flow

Month	Phase	Founder Capital In	Investor In	Revenue In	Outflow	Net	Ending Cash
M1	Pre-Pilot	$8,900	$0	$0	$6,830	$2,070	$2,070
M2	Pre-Pilot	$8,900	$0	$0	$6,830	$2,070	$4,140
M3	Pre-Pilot	$8,900	$0	$0	$6,830	$2,070	$6,210
M4	Pre-Pilot	$8,900	$0	$0	$6,830	$2,070	$8,280
M5	Pre-Pilot	$8,900	$0	$0	$6,830	$2,070	$10,350
M6	Pre-Pilot	$8,900	$0	$0	$7,850	$1,050	$11,400
M7	Pilot 1	$8,900	$0	$400	$10,312	-$1,012	$10,388
M8	Pilot 1	$8,900	$0	$800	$10,312	-$612	$9,776
M9	Pilot 1	$8,900	$0	$1,200	$10,312	-$212	$9,564
M10	Pilot 2	$0	$133,000	$1,800	$12,461	$122,339	$131,903
M11	Pilot 2	$0	$0	$2,200	$12,461	-$10,261	$121,642
M12	Pilot 2	$0	$0	$2,600	$12,461	-$9,861	$111,781
M13	Pilot 2	$0	$0	$3,000	$12,461	-$9,461	$102,320
M14	Pilot 2	$0	$0	$3,400	$12,461	-$9,061	$93,259

M15	Pilot 2	$0	$0	$3,800	$12,461	-$8,661	$84,598
M16	Pilot 2	$0	$0	$4,200	$12,461	-$8,261	$76,337
M17	Pilot 2	$0	$0	$4,600	$12,461	-$7,861	$68,476
M18	Pilot 2	$0	$0	$5,000	$12,461	-$7,461	$61,015
M19	Prod. Readiness	$0	$0	$5,500	$14,942	-$9,442	$51,573
M20	Prod. Readiness	$0	$0	$6,200	$14,942	-$8,742	$42,831
M21	Prod. Readiness	$0	$0	$7,000	$14,942	-$7,942	$34,889
M22	Prod. Readiness	$0	$0	$8,000	$14,942	-$6,942	$27,947
M23	Prod. Readiness	$0	$0	$9,000	$14,942	-$5,942	$22,005
M24	Prod. Readiness	$0	$0	$10,000	$14,942	-$4,942	$17,063
M25	Prod. Readiness	$0	$0	$11,500	$14,942	-$3,442	$13,621
M26	Prod. Readiness	$0	$0	$13,000	$14,942	-$1,942	$11,679
M27	Prod. Readiness	$0	$0	$14,500	$14,942	-$442	$11,237
M28	Prod. Readiness	$0	$0	$16,000	$14,942	$1,058	$12,295
M29	Prod. Readiness	$0	$0	$18,000	$14,942	$3,058	$15,353
M30	Prod. Readiness	$0	$0	$20,000	$14,942	$5,058	$20,411

Totals:

- Founder capital in: $115,700 / INR 9,603,100 (M1-M9 at ~$8,900 x 13 months, adjusted for phase)
- Investor capital in: $133,000 / INR 11,039,000
- Cumulative revenue: ~$171,700
- Cumulative outflow: ~$385,400

ROI and Break-Even

Metric	Value
Total capital invested (founder + investor)	$248,700 / INR 20,642,100
Total economic cost	$385,400 / INR 31,988,200
Sweat-equity gap (economic cost minus cash invested)	~$136,700 / INR 11,346,100
Cumulative revenue at M30	~$171,700 / INR 14,251,100
Monthly break-even (revenue >= outflow)	~M28 (month 28)
Cumulative break-even (total revenue >= total capital)	~M36-M38 (6-8 months post-Production)
Time to ROI (revenue exceeds all economic cost)	~M44-M48 (14-18 months post-Production)

Fundraising, Valuation, and Dilution

Item	Value
Founder/self-funded capital before investor	$115,700 / INR 9,603,100
Investor round size	$133,000 / INR 11,039,000
Recommended structure	SAFE or milestone-based bridge note
Modeled pre-money valuation	$665,000
Modeled post-money valuation	$798,000
Modeled investor dilution from the 20% pool	16.7%

Ownership Pool	Before Round	After Modeled Round	Notes
Owner / founder pool	60%	56.0%	Retained majority control.
Founding + strategic pool	20%	18.7%	Shared pool for founding members and strategic partners.
Investor pool	20% reserved	16.7% issued / 3.3% remaining	Drawn from the dedicated investor pool with minor proportional dilution.

Key Risks

Risk	Likelihood	Impact	Mitigation / Trigger

Investor does not close by M10	High	Critical	Runway runs out by M11 without investor. Pre-negotiate term sheet during Pilot 1. Have backup bridge loan option.
Team fatigue over 30-month timeline	High	High	Maintain clear phase gates. Celebrate milestones. Allow rest periods between phases.
Bootstrapped investor adds insufficient capital	Medium	High	Investor capital barely covers Pilot 2 + Production. Must hit revenue targets by M24 or seek bridge.
Manual ops overload	High	Medium	Add admin capacity before feature expansion. Track callback backlog as release gate.
Regulatory delay (Ethiopia / Kenya)	Medium	High	Continue partner-license model. Count friendly counsel at economic value.
Payment-rail mismatch	Medium	Medium	Use digital invoicing first; localize M-Pesa / Telebirr before scaling.
Security gaps from extended timeline	Medium	Medium	Maintain release freeze discipline. Do not skip hardening gates even under time pressure.
Kenya market entry delayed	Medium	Medium	Focus on Ethiopia stability first. Kenya prep is a Phase 3 activity, not a Phase 2 blocker.

Recommendation

S2-O1-I1 is the **highest-risk, lowest-speed combination** within Scenario 2. The 30-month timeline creates significant fatigue and coordination risk for a part-time team. The bootstrapped investor provides just enough capital to bridge Pilot 2 and early Production, but leaves very thin runway margins (dropping below 1 month at the Pilot 1 close).

This path is viable but fragile. It should only be pursued if:

1. No higher-investment options are available
2. The founding team has strong personal financial resilience to sustain 9 months of self-funding
3. The investor is willing to commit early (term sheet by M7-M8) to avoid a cash crunch at M9-M10
4. Revenue targets in Pilot 2 are treated as hard gates, not aspirational

If the investor close slips past M10 or revenue ramps slower than modeled, this scenario fails. Consider S2-O1-I2 or S2-O1-I3 for meaningful risk reduction.

Assumption Notes

- Android is modeled from the external quote: 480 total hours at $15/hr average across the mobile team.
- Remaining web work is normalized to 150%-160% of Android effort, with an expected case of 745 hours.
- Salary benchmarks for employed coverage: INR 60,000 for doctors ($720/mo), INR 18,000 for admin ops ($217/mo), INR 25,000 for technical support ($301/mo).
- Our L1 (bootstrapped) spend is modeled at ~~40% of ideal team utilization, yielding ~$8,900/mo founder contribution pre-investor (~~$5,930 team + ~$2,970 ops/infra/overhead).
- Investor L1 (bootstrapped) contribution sized at ~$133,000, representing bare-minimum capital to reach Production at constrained pace.
- All INR conversions at 1 USD = 83 INR per params.md.
- FX buffers of 10-15% applied to ETB and KES revenue projections.

End of S2-O1-I1 scenario. All assumptions trace to params.md v2.